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La norma viene de un producto interno si y solo si satisface la identidad del paralelogramo

Teorema 1. *Sea X un espacio vectorial con norma $\|\cdot\|$, entonces*

$$\exists \langle\cdot,\cdot\rangle \text{ producto interior que induce } \|\cdot\| \iff \|\cdot\| \text{ satisface la identidad del paralelogramo.}$$

***Demostración*:** Veamos ambas implicaciones.

$\boxed{\Rightarrow}$ Se sigue de la `identidad-paralelogramo/`**??**.

$\boxed{\Leftarrow}$ Definimos $\forall x, y \in X : \langle x, y\rangle := \frac{1}{4}\left(\|x+y\|^2 - \|x-y\|^2 + i\,\|x+iy\|^2 - i\,\|x-iy\|^2\right)$ para que se satisfaga la identidad de polarización. Veamos que es un producto interno:

(i) Sean $x, y, z \in X$

$$\begin{aligned}
\langle x, z\rangle + \langle y, z\rangle &= \frac{1}{4}\left(\|x+z\|^2 - \|x-z\|^2 + i\,\|x+iz\|^2 - i\,\|x-iz\|^2\right) \\
&\quad + \frac{1}{4}\left(\|y+z\|^2 - \|y-z\|^2 + i\,\|y+iz\|^2 - i\,\|y-iz\|^2\right) \\
&= \frac{1}{4}\Big(\|(x+y)+z\|^2 - \|(x+y)-z\|^2 \\
&\quad + i\,\|(x+y)+iz\|^2 - i\,\|(x+y)-iz\|^2 \Big) = \langle x+y, z\rangle .
\end{aligned}$$

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Referenciado en

- `cor-norma-p-no-prod-interno`
- `teo-l2-unico-esp-hilbert`

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